Optimization problems

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Outline

Optimization problems

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Optimization problems

Introduction: Optimization problems

- In this section we define an **optimization problem**, in minimization and in maximization form, together with its **feasible region** and its **objective function**.
- We define what an **optimal solution** and the **optimal value** are, and we see that an optimal solution may not exist: **unbounded**, **infeasible** problems and **unattainable** infima.
- Finally, we see how every feasible solution gives a **bound** on the optimal value: a simple idea that we will use throughout the course.

Definition 1: optimization problem in minimization and maximization form

Given a set $F \subseteq \mathbb{R}^n$ *and a function* $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$, *with* $F \subseteq D$, *the associated* ***optimization problem*** P *consists of:*

- *in the* ***minimization form***, *finding a point* $\tilde{\boldsymbol{x}}(P) \in F$ *such that*

$$f(\tilde{\boldsymbol{x}}(P)) \leq f(\boldsymbol{x}), \qquad \forall \boldsymbol{x} \in F;$$

- *in the* ***maximization form***, *finding a point* $\tilde{\boldsymbol{x}}(P) \in F$ *such that*

$$f(\tilde{\boldsymbol{x}}(P)) \geq f(\boldsymbol{x}), \qquad \forall \boldsymbol{x} \in F.$$

Feasible region and objective function

- The set F is called the **feasible region** of the optimization problem; its points $\boldsymbol{x} \in F$ are called the **feasible solutions** of the optimization problem.
- The function f is the **objective function** of the optimization problem; it assigns an objective value to each feasible solution.

Among the feasible solutions, we look for the best ones:
the **optimal solutions**.

Definition 2: optimal solution and optimal value (1/2)

- *A point* $\tilde{\boldsymbol{x}}(P) \in F$ *is called an* ***optimal solution*** *of the optimization problem.*
  1. *In the minimization form,*

$$\tilde{\boldsymbol{x}}(P) \in \arg\min\{\, f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \,\}.$$

  2. *In the maximization form,*

$$\tilde{\boldsymbol{x}}(P) \in \arg\max\{\, f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \,\}.$$

Definition 2: optimal solution and optimal value (2/2)

- *The value $f(\tilde{\boldsymbol{x}}(P))$ is called the **optimal value** of the optimization problem, and it is denoted by $z(P)$.*
  1. *In the minimization form,*

$$z(P) = \underbrace{\min\{\, f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \,\}}_{=f(\tilde{\boldsymbol{x}}(P))}.$$

  2. *In the maximization form,*

$$z(P) = \underbrace{\max\{\, f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \,\}}_{=f(\tilde{\boldsymbol{x}}(P))}.$$

Notation

- An optimization problem may admit multiple (possibly infinitely many) optimal solutions, or none at all. In such cases, the operators arg min and arg max return the full set of optimal solutions when they exist, and return the empty set otherwise.
- With a slight abuse of notation:
  1. in the **minimization form**, we write the optimization problem as
  $$\min\{ f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \};$$
  2. in the **maximization form**, we write the optimization problem as
  $$\max\{ f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \}.$$

The goal of an optimization problem

Although we use the symbols "min" and "max", the goal of an optimization problem is to **find an optimal solution** $\tilde{x}(P)$, whenever it exists, rather than merely determining the optimal value $z(P)$.

How to read the figures

- In the figures of this chapter:
  - the graph of the objective function over the feasible region is red;
  - the feasible region is the red part of the x-axis;
  - the optimal solutions are blue;
  - an empty circle marks an endpoint that does not belong to the feasible region.

- We now look at some examples: the optimal solutions are found directly from the definition.

Example: a quadratic objective function (1/3)

Consider the following objective function and feasible region:

$$f : \mathbb{R} \to \mathbb{R},\ f : x \mapsto x^2 \qquad \text{and} \qquad F = \{\, x \in \mathbb{R} : -1 \leq x \leq 2 \,\}.$$

The feasible region is the closed interval $F = [-1, 2]$.

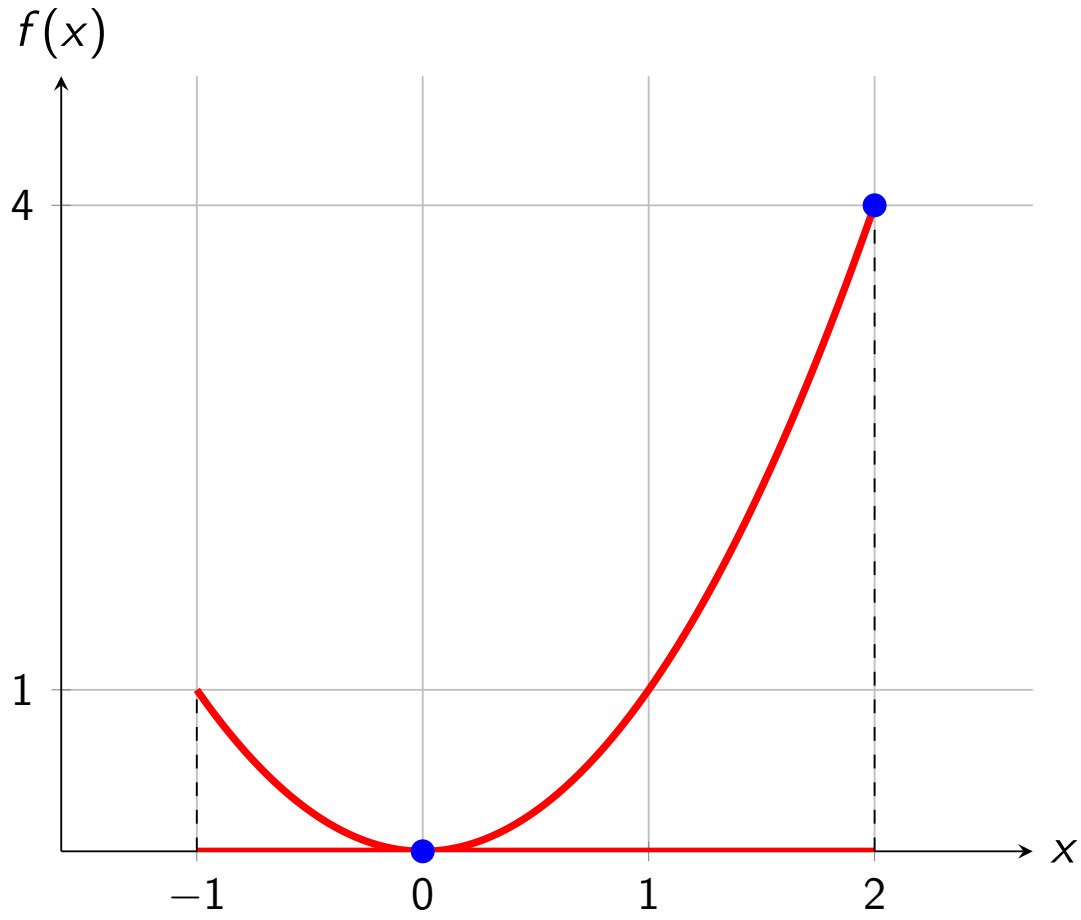
The associated optimization problems in minimization and maximization forms are:

$$\min \left\{ x^2 \ \text{ subject to } \ -1 \leq x \leq 2 \right\}$$

and

$$\max \left\{ x^2 \ \text{ subject to } \ -1 \leq x \leq 2 \right\}$$

Example: a quadratic objective function (2/3)



Example: a quadratic objective function (3/3)

- For the minimization form, since $f(x) = x^2 \geq 0 = f(0)$ for every feasible x, and $x^2 = 0$ only for $x = 0$, the unique optimal solution and the optimal value are:

$$\tilde{\boldsymbol{x}}(P) = (0) \quad \text{and} \quad z(P) = 0.$$

- For the maximization form, for every feasible x we have $|x| \leq 2$ and then $f(x) = |x|^2 \leq 4 = f(2)$, with $x^2 = 4$ only for $x = -2$ and $x = 2$; since $-2 \notin F$, the unique optimal solution and the optimal value are:

$$\tilde{\boldsymbol{x}}(P) = (2) \quad \text{and} \quad z(P) = 4.$$

The optimal solutions are found directly from the definition:

we compare the value of the objective function at a candidate point with its value at every feasible solution.

In the next example the optimal solutions are infinitely many.

Example: infinitely many optimal solutions (1/3)

Consider the following objective function and feasible region:

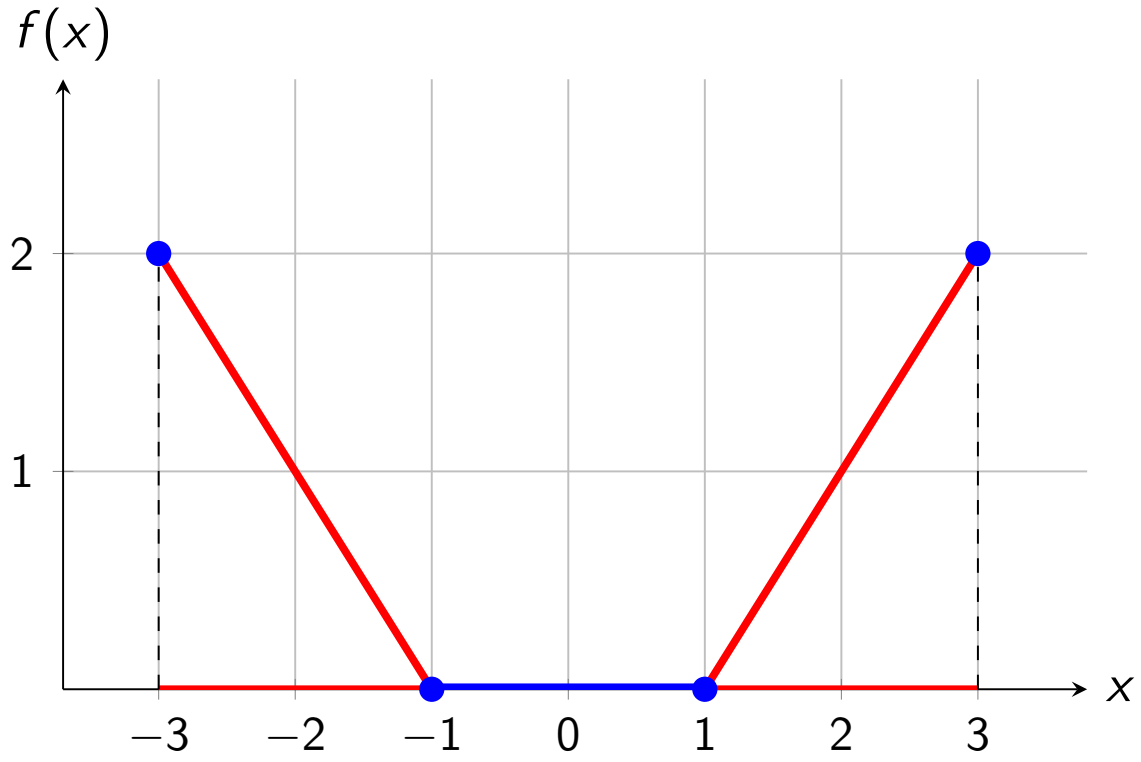
$$f : \mathbb{R} \to \mathbb{R},\ f : x \mapsto \max\{\, 0,\ |x| - 1 \,\} \qquad \text{and} \qquad F = [-3, 3].$$

The objective function is

$$f(x) = 0 \ \text{ for } \ -1 \leq x \leq 1 \qquad \text{and} \qquad f(x) = |x| - 1 \ \text{ otherwise :}$$

its graph is made of three segments.

Example: infinitely many optimal solutions (2/3)



Example: infinitely many optimal solutions (3/3)

- For the minimization form, $f(x) \geq 0$ for every feasible x, and $f(x) = 0$ exactly for $-1 \leq x \leq 1$. The problem has **infinitely many** optimal solutions (the blue segment) and the optimal value is:

  $$\arg\min \left\{ \max\{0, |x| - 1\} \text{ subject to } -3 \leq x \leq 3 \right\} = [-1, 1]$$

  and $z(P) = 0$.

- For the maximization form, for every feasible x we have $|x| \leq 3$ and then $f(x) \leq \max\{0, 3 - 1\} = 2$, with $f(x) = 2$ only for $x = -3$ and $x = 3$. The problem has **two** optimal solutions and the optimal value is:

  $$\arg\max \left\{ \max\{0, |x| - 1\} \text{ subject to } -3 \leq x \leq 3 \right\} = \{-3, 3\}$$

  and $z(P) = 2$.

When an optimal solution does not exist (1/2)

An optimal solution of an optimization problem may not exist for one of the following three reasons:

1. If the objective function decreases indefinitely in a minimization problem or increases indefinitely in a maximization problem, the optimization problem is said to be **unbounded**. In these cases, $z(P) = -\infty$ and $z(P) = +\infty$, respectively.

2. If the feasible region is empty, meaning that no solutions exist, the optimization problem is said to be **infeasible**. In these cases, $z(P)$ is not defined.

When an optimal solution does not exist (2/2)

3. If the infimum (or supremum) of the objective function exists but is not attained within the feasible region, the optimization problem is said to have an **unattainable infimum (or supremum)**. In these cases, $z(P)$ is not defined.

- In the **minimization form**, an optimal solution is a **point of minimum**, and the corresponding optimal value is the **minimum** of the objective function over the feasible region.
- Similarly, in the **maximization form**, an optimal solution is a **point of maximum**, and the corresponding optimal value is the **maximum** of the objective function over the feasible region.

We now see one example for each case:

an **unbounded** problem, an **infeasible** problem
and two problems with an **unattainable infimum**.

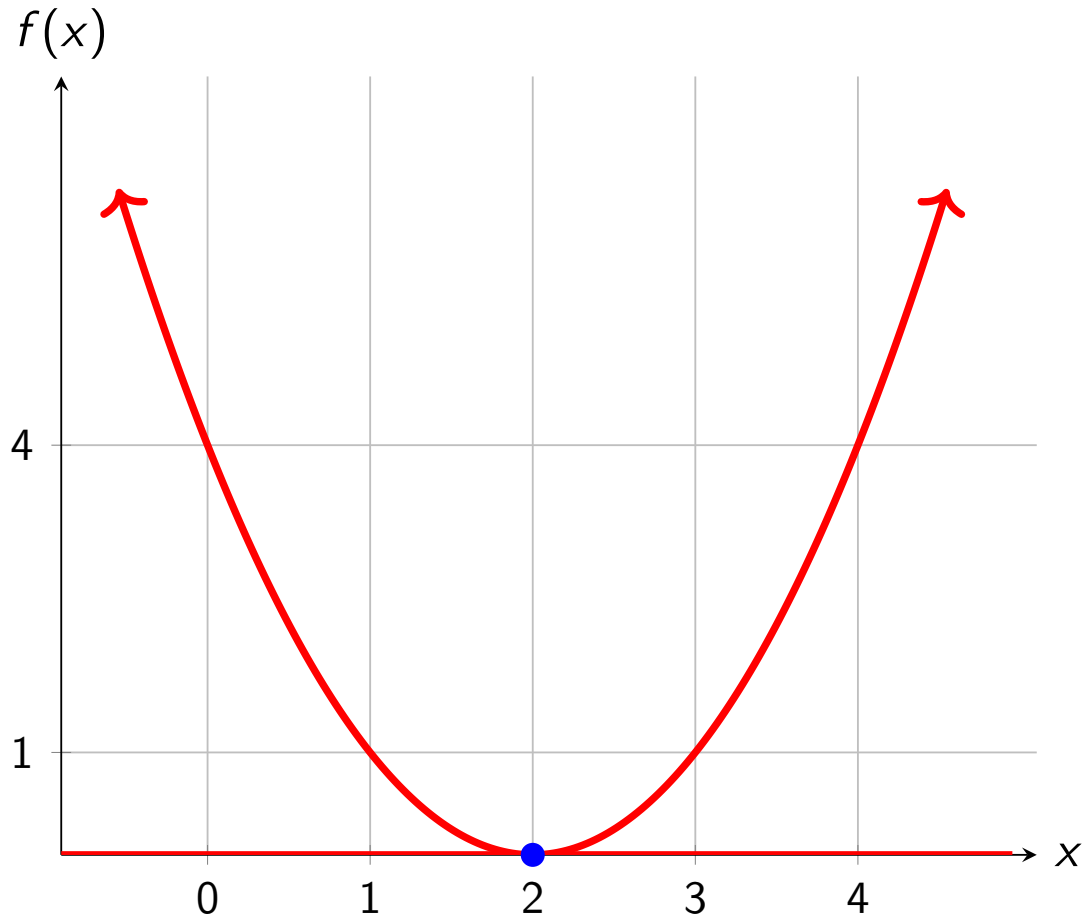
Example: an unbounded problem (1/3)

Consider the following objective function and feasible region:

$$f : \mathbb{R} \to \mathbb{R}, \; f : x \mapsto (x-2)^2 \qquad \text{and} \qquad F = \mathbb{R}.$$

The feasible region is the whole real line: the graph continues upward on both sides of the figure.

Example: an unbounded problem (2/3)



Example: an unbounded problem (3/3)

- For the minimization form, since $f(x) = (x-2)^2 \geq 0 = f(2)$ for every $x \in \mathbb{R}$, and $(x-2)^2 = 0$ only for $x = 2$, the unique optimal solution and the optimal value are:

$$\tilde{\boldsymbol{x}}(P) = (2) \quad \text{and} \quad z(P) = 0.$$

- For the maximization form, the objective function increases indefinitely: for every value $M \geq 0$, the feasible solution $x = M + 3$ has

$$f(M+3) = (M+1)^2 \geq M + 1 > M.$$

No feasible solution can be optimal and the problem is **unbounded**: $z(P) = +\infty$.

Example: an infeasible problem

Consider the following objective function and feasible region:

$$f : \mathbb{R} \to \mathbb{R},\ f : x \mapsto x^2 \qquad \text{and} \qquad F = \{\, x \in \mathbb{R} : x^2 \leq -1 \,\}.$$

- Since $x^2 \geq 0 > -1$ for every $x \in \mathbb{R}$, no point satisfies the constraint and the feasible region is empty: $F = \emptyset$.
- Both the minimization and the maximization form are **infeasible**: they have no feasible solution, hence no optimal solution, and $z(P)$ is not defined.

Example: an unattainable infimum (1/4)

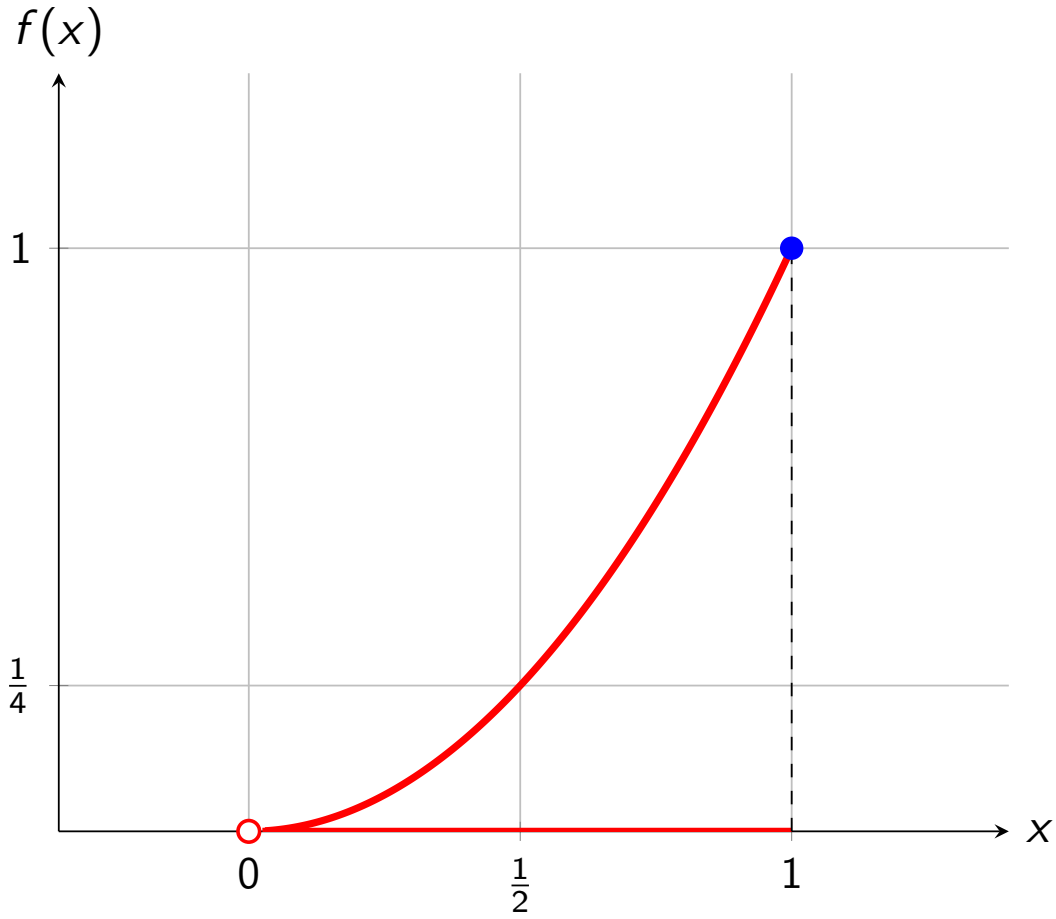
Consider the following objective function and feasible region:

$$f : \mathbb{R} \to \mathbb{R},\ f : x \mapsto x^2 \qquad \text{and} \qquad F = \{\, x \in \mathbb{R} : 0 < x \leq 1 \,\}.$$

The feasible region is the interval $F = (0, 1]$, open on the left.

The point $x = 0$ does not belong to the feasible region (empty circle in the figure).

Example: an unattainable infimum (2/4)



Example: an unattainable infimum (3/4)

- For the minimization form, $f(x) = x^2 > 0$ for every feasible x, and the values of f get as close to 0 as we want: the infimum of f over F is 0. However, no feasible solution is optimal: for every $x \in F$, the point $\frac{x}{2}$ is feasible and

$$f\left(\frac{x}{2}\right) = \frac{x^2}{4} < x^2 = f(x).$$

  The problem has an **unattainable infimum**: there is no optimal solution and $z(P)$ is not defined.

Example: an unattainable infimum (4/4)

- For the maximization form, for every feasible x we have $0 < x \leq 1$ and then $f(x) = x^2 \leq 1 = f(1)$, with $x^2 = 1$ only for $x = 1$ in F. The unique optimal solution and the optimal value are:

$$\tilde{\boldsymbol{x}}(P) = (1) \quad \text{and} \quad z(P) = 1.$$

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In the previous example the infimum is approached at an endpoint of the feasible region that does not belong to it.

In the next example the infimum is approached **at infinity**.

Example: an unattainable infimum at infinity (1/4)

Consider the following objective function and feasible region:

$$f : \mathbb{R} \setminus \{0\} \to \mathbb{R},\ f : x \mapsto \frac{1}{x} \qquad \text{and} \qquad F = \{\, x \in \mathbb{R} : x > 0 \,\}.$$

The feasible region is the open interval $F = (0, +\infty)$.

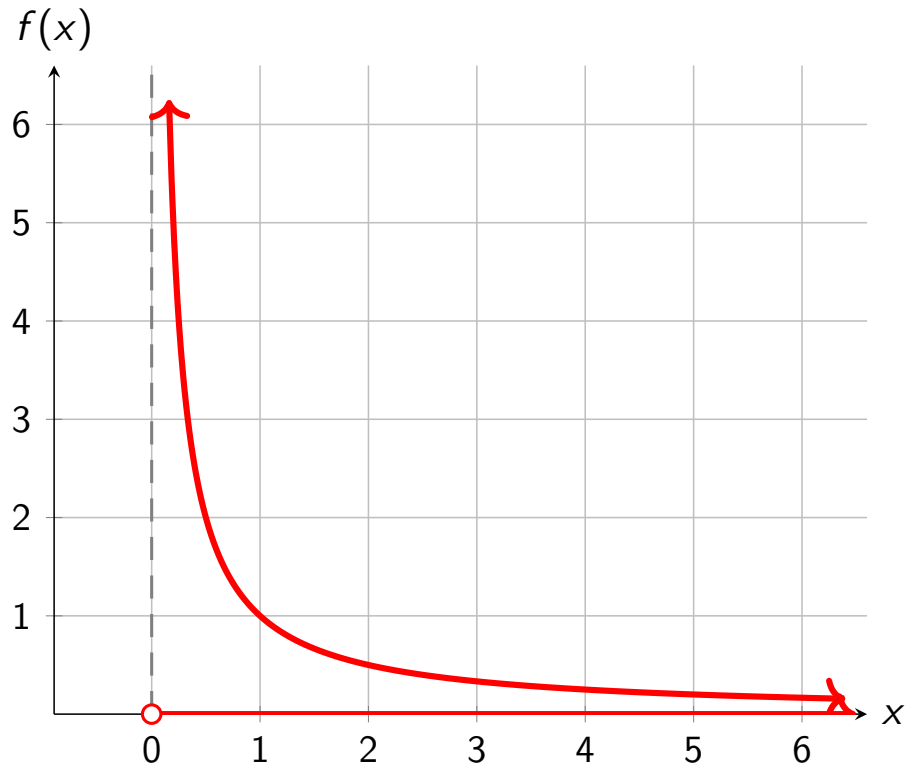
The associated optimization problems in minimization and maximization forms are:

$$\min \left\{ \frac{1}{x} \quad \text{subject to } x > 0 \right\}$$

and

$$\max \left\{ \frac{1}{x} \quad \text{subject to } x > 0 \right\}$$

Example: an unattainable infimum at infinity (2/4)



The point $x = 0$ does not belong to the feasible region (empty circle). The graph continues on the right, approaching the x-axis, and upward, approaching the dashed line $x = 0$.

Example: an unattainable infimum at infinity (3/4)

For the minimization form, $f(x) = \frac{1}{x} > 0$ for every feasible x, and the values of f get as close to 0 as we want as $x \to +\infty$: the infimum of f over F is 0.

However, no feasible solution is optimal: for every $x \in F$, the point $2x$ is feasible and

$$f(2x) = \frac{1}{2x} < \frac{1}{x} = f(x).$$

The problem has an **unattainable infimum**: there is no optimal solution and

$$\inf \left\{ \frac{1}{x} \text{ subject to } x > 0 \right\} = 0, \qquad z(P) \text{ is not defined.}$$

Example: an unattainable infimum at infinity (4/4)

For the maximization form, the objective function increases indefinitely as $x \to 0^+$: for every value $M > 0$, the feasible solution $x = \frac{1}{M+1}$ has

$$f\left(\frac{1}{M+1}\right) = M + 1 > M.$$

No feasible solution can be optimal and the problem is **unbounded**:

$$\sup\left\{ \frac{1}{x} \quad \text{subject to } x > 0 \right\} = +\infty \qquad \text{and} \qquad z(P) = +\infty.$$

Even without knowing an optimal solution, every **feasible solution** tells us something about the optimal value $z(P)$.

Feasible solutions and bounds on the optimal value (1/2)

Given

$$\underbrace{\begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_n \end{pmatrix}}_{\bar{\boldsymbol{x}}(P)} \in F$$

a **feasible solution** of an optimization problem, we have

1. in **minimization** form, the following **upper bound**:

$$z(P) \leq \underbrace{f(\bar{\boldsymbol{x}}(P))}_{UB(\bar{\boldsymbol{x}})};$$

Feasible solutions and bounds on the optimal value (2/2)

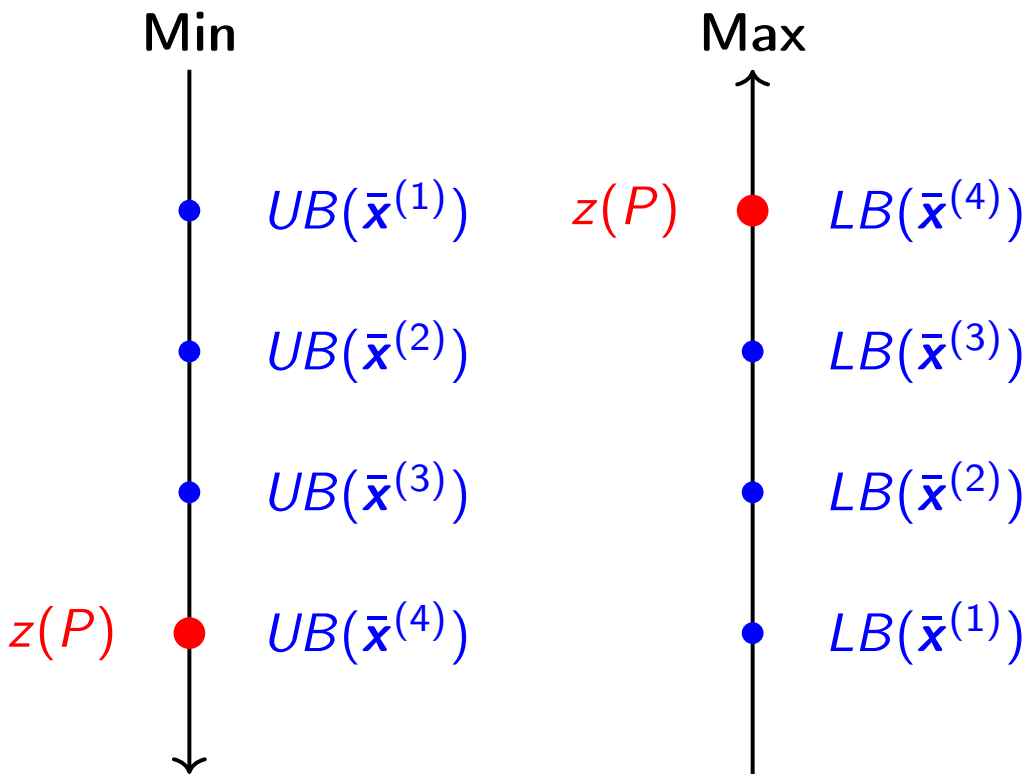
Given a **feasible solution** $\bar{\boldsymbol{x}}(P) \in F$ of an optimization problem, we have

2. in **maximization** form, the following **lower bound**:

$$\underbrace{f(\bar{\boldsymbol{x}}(P))}_{LB(\bar{\boldsymbol{x}})} \leq z(P).$$

Upper and lower bounds

Four feasible solutions $\bar{\boldsymbol{x}}^{(1)}, \bar{\boldsymbol{x}}^{(2)}, \bar{\boldsymbol{x}}^{(3)}, \bar{\boldsymbol{x}}^{(4)} \in F$ provide upper bounds in the minimization case and lower bounds in the maximization case for the optimal value $z(P)$ (the fourth one attains the optimal value).



Example: bounds from feasible solutions (1/2)

Consider again the objective function $f(x) = x^2$ and the feasible region $F = [-1, 2]$ of the example with a quadratic objective function, where $z(P) = 0$ in minimization form and $z(P) = 4$ in maximization form.

The feasible solutions

$$\bar{\boldsymbol{x}}^{(1)} = (-1), \qquad \bar{\boldsymbol{x}}^{(2)} = \left(\frac{1}{2}\right), \qquad \bar{\boldsymbol{x}}^{(3)} = (1)$$

have objective values 1, $\frac{1}{4}$ and 1.

Example: bounds from feasible solutions (2/2)

- In minimization form, they give the upper bounds

$$UB(\bar{\boldsymbol{x}}^{(1)}) = 1, \qquad UB(\bar{\boldsymbol{x}}^{(2)}) = \frac{1}{4}, \qquad UB(\bar{\boldsymbol{x}}^{(3)}) = 1$$

  on $z(P) = 0$; the best one is $\frac{1}{4}$, the smallest.
- In maximization form, they give the lower bounds

$$LB(\bar{\boldsymbol{x}}^{(1)}) = 1, \qquad LB(\bar{\boldsymbol{x}}^{(2)}) = \frac{1}{4}, \qquad LB(\bar{\boldsymbol{x}}^{(3)}) = 1$$

  on $z(P) = 4$; the best one is 1, the largest.

None of these bounds is equal to the optimal value: a bound tells us how far the optimal value can be from the value of a known feasible solution, without computing it.

Summary (1/2)

Optimization problem:

- Feasible region F, objective function f, minimization or maximization form
- Optimal solution $\tilde{\boldsymbol{x}}(P)$ and optimal value $z(P) = f(\tilde{\boldsymbol{x}}(P))$
- The optimal solutions may be one, several, infinitely many, or none

The goal:

- Find an optimal solution, not only the optimal value

Summary (2/2)

When an optimal solution does not exist:

- **Unbounded**: $z(P) = -\infty$ (min) or $z(P) = +\infty$ (max)
- **Infeasible**: $F = \emptyset$, $z(P)$ not defined
- **Unattainable infimum (supremum)**: at an open endpoint, as for x^2 on $(0, 1]$, or at infinity, as for $\frac{1}{x}$ on $(0, +\infty)$

Bounds:

- Every feasible solution gives an upper bound (min) or a lower bound (max) on $z(P)$

Properties of optimization problems

Introduction: Properties of optimization problems

- In this section we see four simple properties that hold for **every** optimization problem.
- They tell us how the optimal value changes when we change the sign of the objective function, add a constant to it, or multiply it by a positive or a negative constant.
- They allow us to rewrite a problem in the most convenient form: for instance, every maximization problem can be written as a minimization problem.

Before we start

- The following properties hold for every optimization problem.
- We state the properties using min and max only in the case where the minimum and maximum exist (i.e., they are attained on the feasible region); otherwise, the analogous statements hold by replacing min / max with inf / sup.
- In the figures of this section:
  - the original objective function is red, the transformed one is violet;
  - the feasible region is the red part of the x-axis;
  - the optimal solutions are blue.

Property: minimum and maximum

- For every objective function f and every feasible region F, we have:

$$\begin{aligned} \min \left\{ f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \right\} \\ = - \max \left\{ -f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \right\} \end{aligned} \tag{1}$$

- The optimal solutions do not change: a point minimizes f over F if and only if it maximizes $-f$ over F.

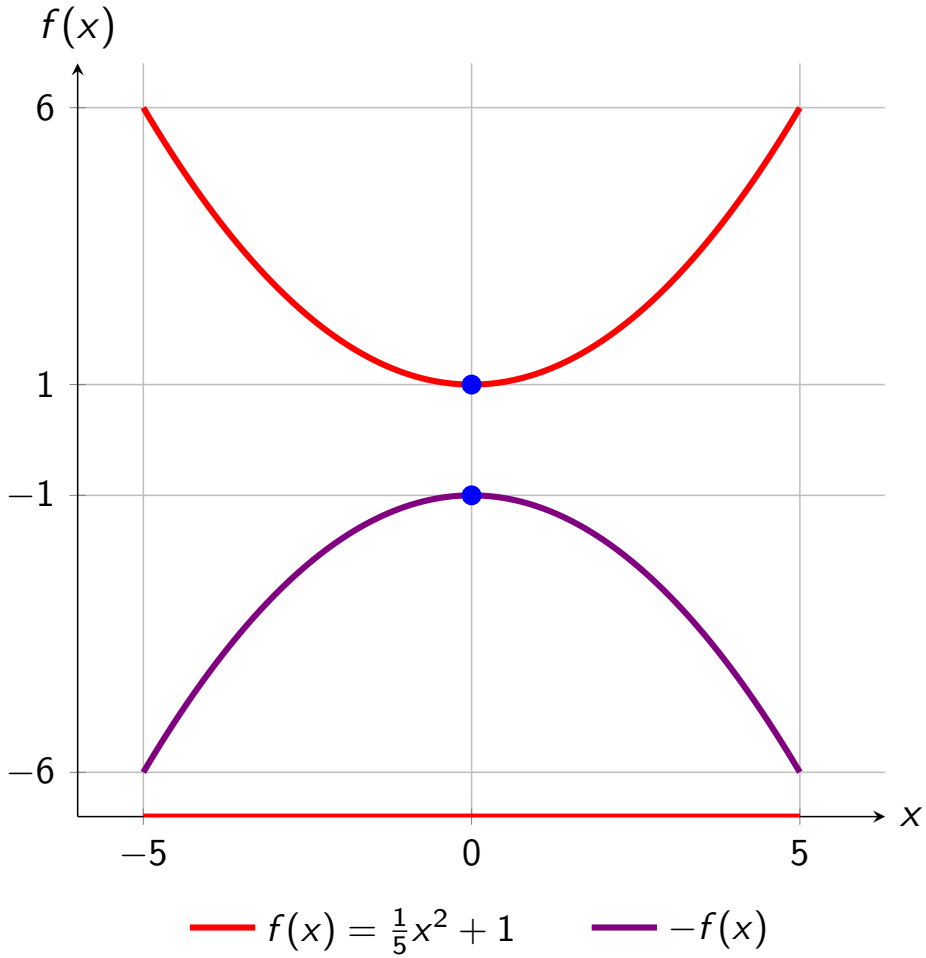
Example: minimum and maximum (1/2)

$$\underbrace{\min\left\{\frac{1}{5}x^2+1 \text{ subject to } -5\le x\le 5\right\}}_{=1}$$

$$= -\underbrace{\max\left\{-\left(\frac{1}{5}x^2+1\right) \text{ subject to } -5\le x\le 5\right\}}_{=-1}$$

Indeed, $\frac{1}{5}x^2+1\ge 1$ and $-\left(\frac{1}{5}x^2+1\right)\le -1$ for every x, with equality only for $x=0$: both optima are attained at $x=0$.

Example: minimum and maximum (2/2)



Property: constant addition

- For every objective function, every feasible region, and every $a \in \mathbb{R}$, we have:

$$\begin{aligned} &\min \left\{ a + f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \right\} \\ &\qquad = a + \min \left\{ f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \right\} \end{aligned} \tag{2}$$

- Adding a constant moves the graph of the objective function up or down: the optimal solutions do not change.

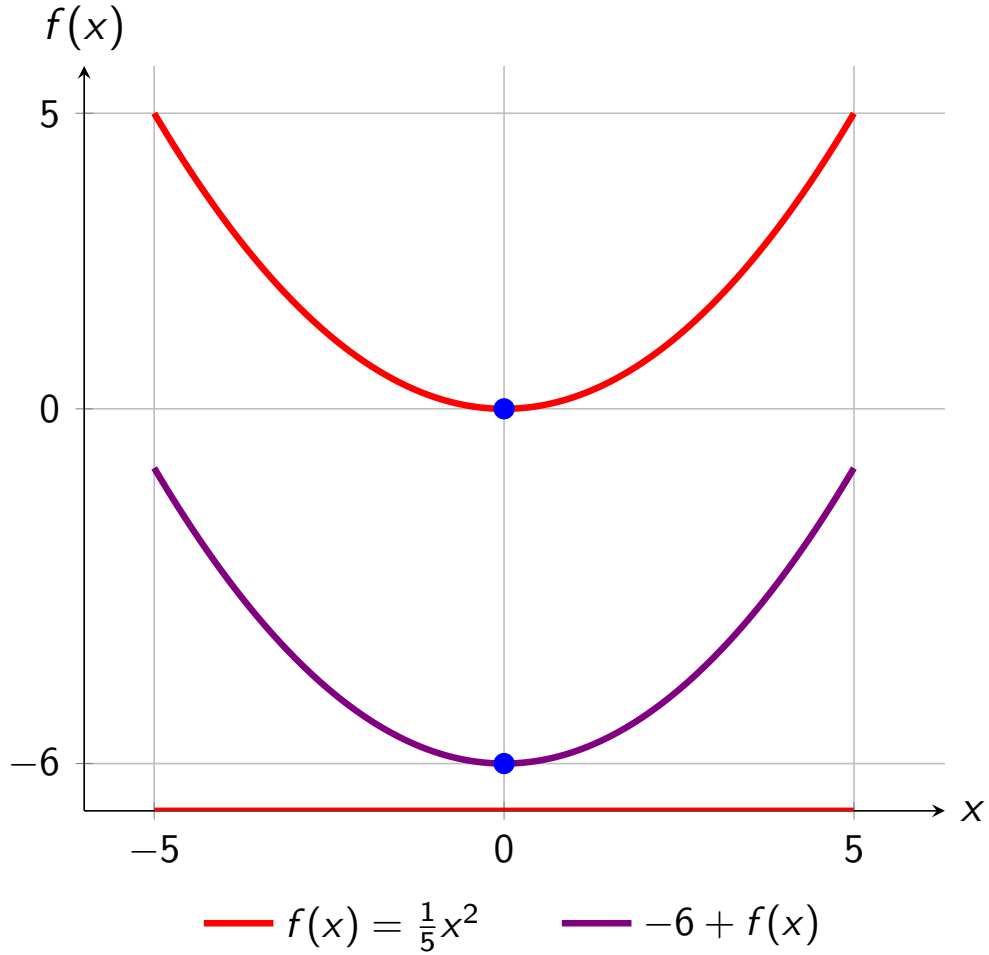
Example: constant addition (1/2)

$$\underbrace{\min\left\{\frac{1}{5}x^2 - 6 \text{ subject to } -5 \leq x \leq 5\right\}}_{=-6}$$

$$= -6 + \underbrace{\min\left\{\frac{1}{5}x^2 \text{ subject to } -5 \leq x \leq 5\right\}}_{=0}$$

Both minima are attained at $x = 0$, since $\frac{1}{5}x^2 \geq 0$ for every x, with equality only for $x = 0$.

Example: constant addition (2/2)



Property: multiplication by a positive constant

- For every objective function, every feasible region, and every $a \in \mathbb{R}_{>0}$, we have:

$$\begin{aligned} &\min \left\{ a\, f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \right\} \\ &\qquad = a \ \min \left\{ f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \right\} \end{aligned} \tag{3}$$

- Multiplying by a positive constant stretches (or compresses) the graph vertically: the optimal solutions do not change.

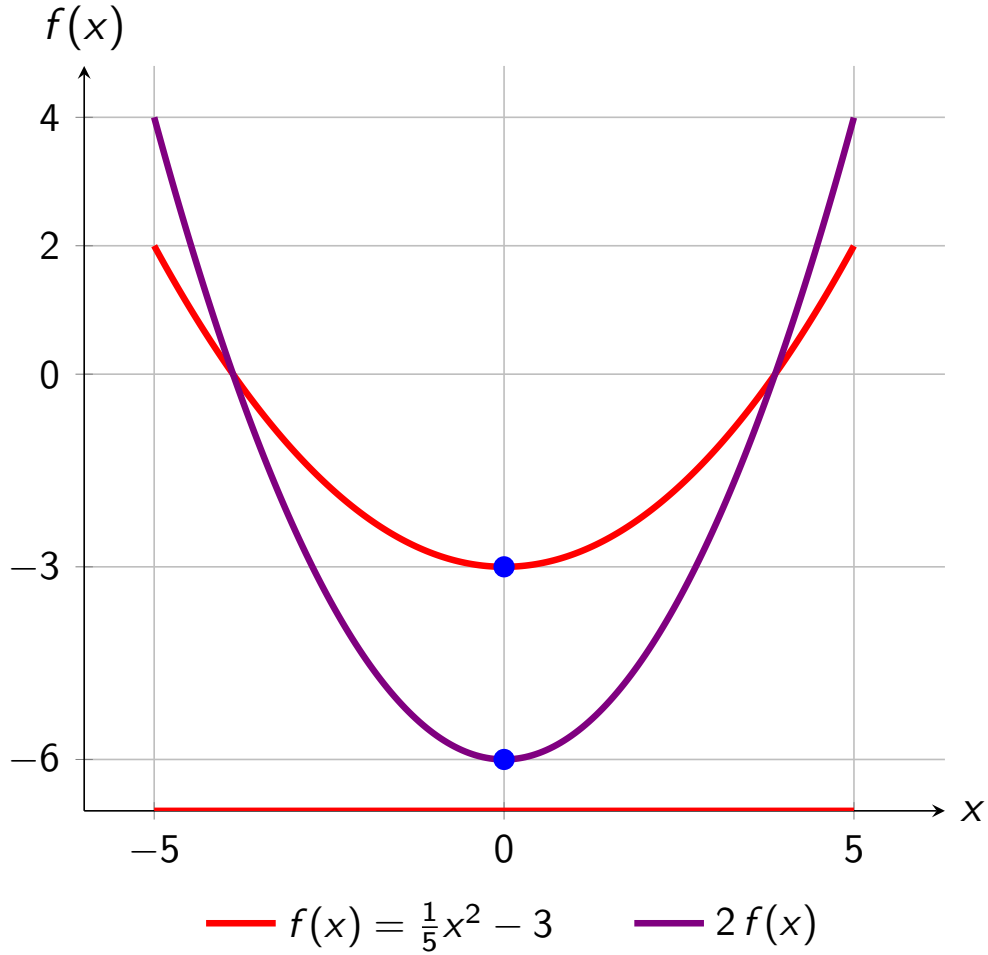
Example: multiplication by a positive constant (1/2)

$$\underbrace{\min\left\{ 2\left(\frac{1}{5}x^2 - 3\right) \text{ subject to } -5 \leq x \leq 5 \right\}}_{=-6}$$

$$= 2 \underbrace{\min\left\{ \frac{1}{5}x^2 - 3 \text{ subject to } -5 \leq x \leq 5 \right\}}_{=-3}$$

Both minima are attained at $x = 0$, since $\frac{1}{5}x^2 - 3 \geq -3$ for every x, with equality only for $x = 0$.

Example: multiplication by a positive constant (2/2)



Property: multiplication by a negative constant

- For every objective function, every feasible region, and every $a \in \mathbb{R}_{<0}$, we have:

$$\begin{aligned} \min \left\{ a\, f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \right\} \\ = a \, \max \left\{ f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \right\} \end{aligned} \tag{4}$$

- Multiplying by a negative constant turns the graph upside down: the points of minimum of $a\, f$ are the points of maximum of f.

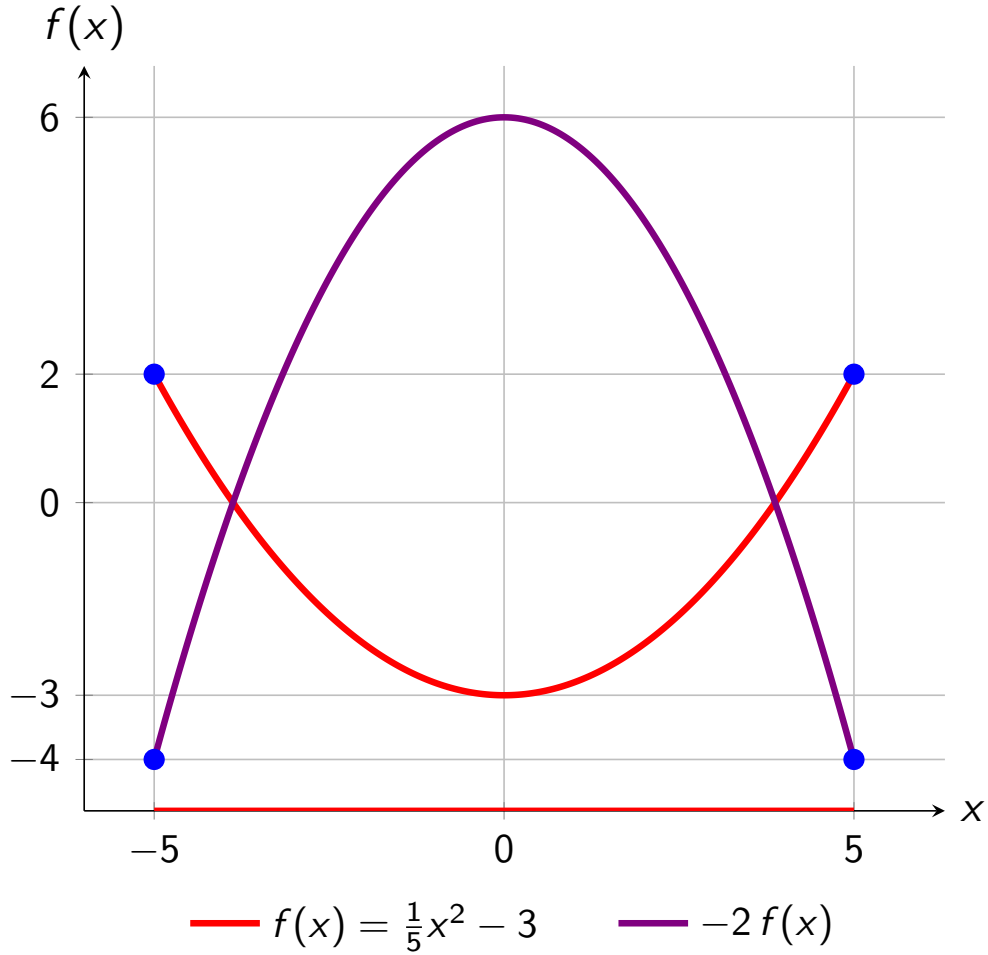
Example: multiplication by a negative constant (1/2)

$$\underbrace{\min\left\{ -2\left(\frac{1}{5}x^2 - 3\right) \text{ subject to } -5 \leq x \leq 5 \right\}}_{=-4}$$

$$= -2 \underbrace{\max\left\{ \frac{1}{5}x^2 - 3 \text{ subject to } -5 \leq x \leq 5 \right\}}_{=2}$$

For every feasible x we have $|x| \leq 5$ and then $\frac{1}{5}x^2 - 3 \leq \frac{25}{5} - 3 = 2$, with equality only for $x = -5$ and $x = 5$: both problems have the **two** optimal solutions $x = -5$ and $x = 5$.

Example: multiplication by a negative constant (2/2)



Summary

Properties of every optimization problem (with inf / sup when the optimum is not attained):

- $\min\{f\} = -\max\{-f\}$: a point minimizes f if and only if it maximizes $-f$
- $\min\{a + f\} = a + \min\{f\}$ for every $a \in \mathbb{R}$
- $\min\{a\, f\} = a \min\{f\}$ for every $a > 0$
- $\min\{a\, f\} = a \max\{f\}$ for every $a < 0$

Optimal solutions:

- Adding a constant or multiplying by a positive constant does not change them
- Multiplying by a negative constant exchanges points of minimum and points of maximum

Local and global optimal solutions

Introduction: Local and global optimal solutions

- In this section we define the points of **minimum** and **maximum** of a function: the optimal solutions are the **global optima**.
- We introduce the **spherical** δ-**neighborhoods** and, with them, the points of **local** minimum and maximum.
- Many methods only look at the points close to a given point: they can stop at a local optimum that is not global. Knowing the difference is fundamental.

Definition 3: point of minimum/maximum and minimum/maximum

Given a function $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$, *a point* $\tilde{\boldsymbol{x}} \in D$ *is*

- *a* ***point of minimum (global minimum point)*** *of* f *in* D *if*

$$f(\tilde{\boldsymbol{x}}) \leq f(\boldsymbol{x}), \quad \forall \boldsymbol{x} \in D$$

  and $f(\tilde{\boldsymbol{x}})$ *is the* ***minimum*** *of* f *in* D*;*

- *a* ***point of maximum (global maximum point)*** *of* f *in* D *if*

$$f(\tilde{\boldsymbol{x}}) \geq f(\boldsymbol{x}), \quad \forall \boldsymbol{x} \in D$$

  and $f(\tilde{\boldsymbol{x}})$ *is the* ***maximum*** *of* f *in* D*.*

Global optima

- A point of minimum (maximum) is also called a **minimizer** (**maximizer**) of f in D.
- There can exist multiple points of minimum (maximum), possibly infinitely many, as in the example with infinitely many optimal solutions.
- The optimal solutions of a minimization (maximization) problem with objective function f and feasible region F are the points of minimum (maximum) of f in F: they are its **global optima**.

To define the **local** optima, we first need the notion of neighborhood of a point.

Definition 4: spherical δ-neighborhood

Given a point $\bar{\boldsymbol{x}} \in \mathbb{R}^n$ *and a value* $\delta \in \mathbb{R}_{>0}$, *the associated* ***spherical*** δ***-neighborhood*** *is:*

$$N(\bar{\boldsymbol{x}}, \delta) = \left\{ \boldsymbol{x} \in \mathbb{R}^n : \ ||\bar{\boldsymbol{x}} - \boldsymbol{x}|| < \delta \right\}$$

- The value $||\bar{\boldsymbol{x}} - \boldsymbol{x}||$ is the **Euclidean distance** between a generic point $\boldsymbol{x} \in \mathbb{R}^n$ and the point $\bar{\boldsymbol{x}} \in \mathbb{R}^n$, the **center** of the neighborhood:

$$||\bar{\boldsymbol{x}} - \boldsymbol{x}|| = \sqrt{\sum_{j=1}^{n} (\bar{x}_j - x_j)^2}.$$

Spherical neighborhoods in $\mathbb{R}$ and in $\mathbb{R}^2$

- The neighborhood $N(\bar{\boldsymbol{x}}, \delta)$ contains the points whose distance from the center is smaller than the **radius** δ.
- In $\mathbb{R}$ it is the open interval $(\bar{x} - \delta,\ \bar{x} + \delta)$.
- In $\mathbb{R}^2$ it is the open disk of center $\bar{\boldsymbol{x}}$ and radius δ, without its boundary circle.

Example: spherical neighborhoods (1/3)

Consider the point $\bar{\boldsymbol{x}} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} \in \mathbb{R}^2$ and the radii $\delta_1 = 1$ and $\delta_2 = 2$.

The associated spherical δ-neighborhoods are

$$N\left(\begin{pmatrix} 4 \\ 4 \end{pmatrix}, 1\right) = \left\{ \boldsymbol{x} \in \mathbb{R}^2 : \sqrt{(4 - x_1)^2 + (4 - x_2)^2} < 1 \right\}$$

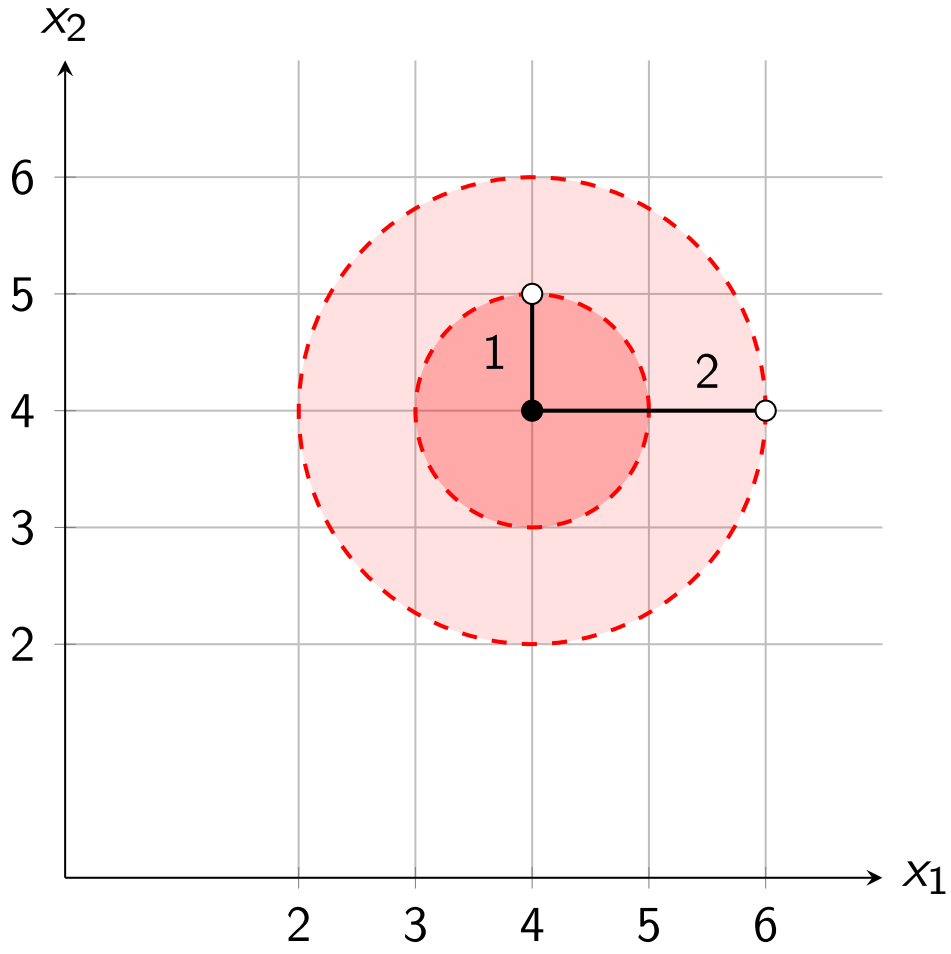
and

$$N\left(\begin{pmatrix} 4 \\ 4 \end{pmatrix}, 2\right) = \left\{ \boldsymbol{x} \in \mathbb{R}^2 : \sqrt{(4 - x_1)^2 + (4 - x_2)^2} < 2 \right\}.$$

Example: spherical neighborhoods (2/3)

- The two neighborhoods are the red disks in the figure of the next slide: the darker disk is $N(\bar{\boldsymbol{x}}, 1)$, which is contained in $N(\bar{\boldsymbol{x}}, 2)$.
- The boundary circles are dashed because they do not belong to the neighborhoods.
- For instance, the points $\begin{pmatrix} 4 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} 6 \\ 4 \end{pmatrix}$ (empty circles) have distance exactly 1 and 2 from the center, respectively.

Example: spherical neighborhoods (3/3)



With the spherical neighborhoods we can now compare a point only with the points **close to it**:

this gives the points of **local** minimum and maximum.

Definition 5: point of local minimum/maximum and local minimum/maximum

Given a function $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$, *a point* $\bar{\boldsymbol{x}} \in D$ *is*

- *a* ***point of local minimum*** *of* f *in* D *if there exists a spherical neighborhood* $N(\bar{\boldsymbol{x}}, \delta)$ *with* $\delta \in \mathbb{R}_{>0}$ *such that:*

$$f(\bar{\boldsymbol{x}}) \leq f(\boldsymbol{x}), \quad \forall \boldsymbol{x} \in N(\bar{\boldsymbol{x}}, \delta) \cap D$$

  and $f(\bar{\boldsymbol{x}})$ *is a* ***local minimum*** *of* f *in* D*;*

- *a* ***point of local maximum*** *of* f *in* D *if there exists a spherical neighborhood* $N(\bar{\boldsymbol{x}}, \delta)$ *with* $\delta \in \mathbb{R}_{>0}$ *such that:*

$$f(\bar{\boldsymbol{x}}) \geq f(\boldsymbol{x}), \quad \forall \boldsymbol{x} \in N(\bar{\boldsymbol{x}}, \delta) \cap D$$

  and $f(\bar{\boldsymbol{x}})$ *is a* ***local maximum*** *of* f *in* D*.*

Local optima

- There can exist multiple points of local minimum (maximum), possibly infinitely many.
- To prove that a point is a point of local minimum (maximum) with the definition, it is enough to exhibit **one** radius δ for which the inequality holds in $N(\bar{\boldsymbol{x}}, \delta) \cap D$; any smaller radius also works.

Example: points of local minimum (1/4)

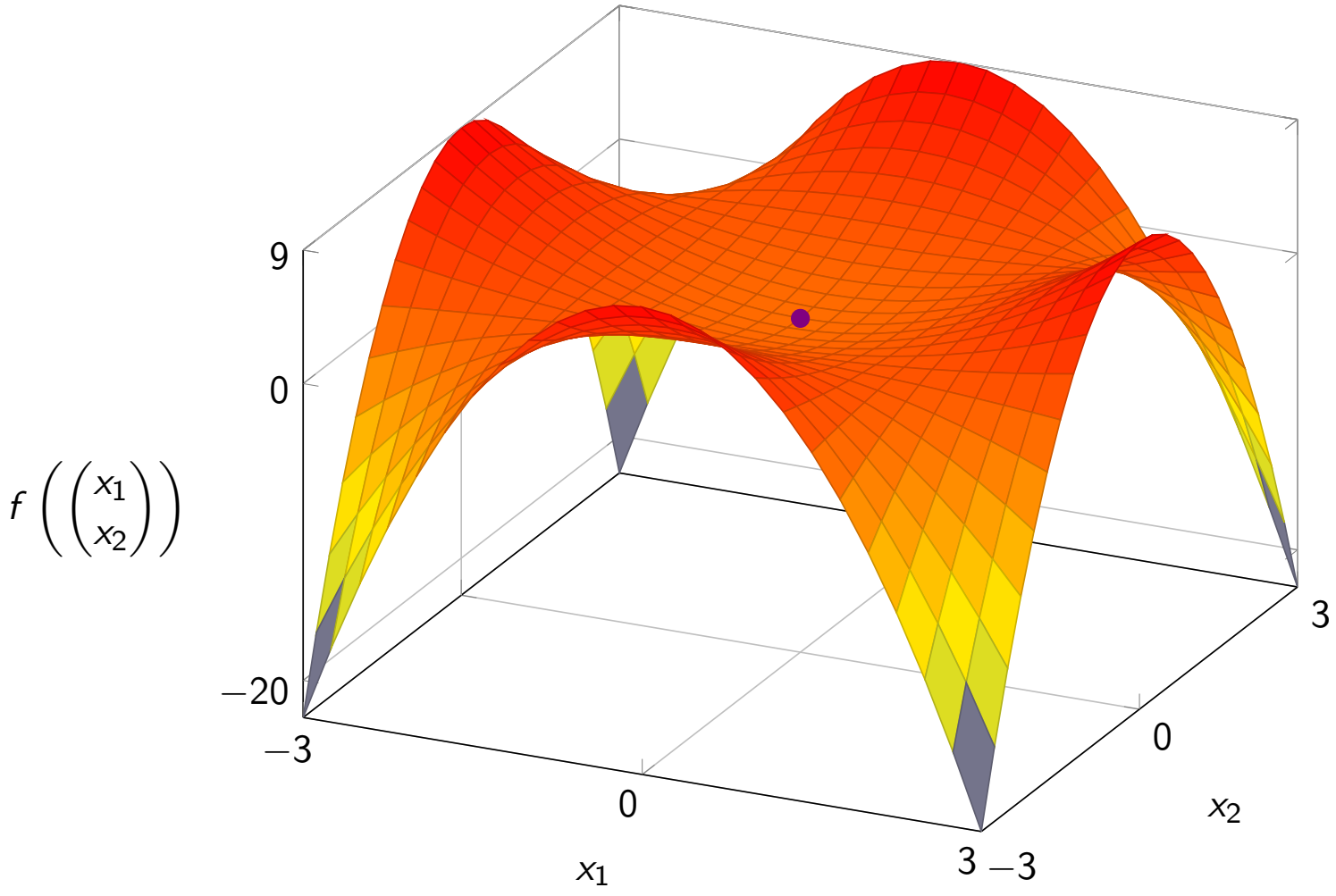
Consider the following function:

$$f : \mathbb{R}^2 \to \mathbb{R}, \quad f : \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mapsto x_1^2 + x_2^2 - \frac{1}{2} x_1^2 x_2^2.$$

In the next slide we see the graph of the function in $[-3, 3] \times [-3, 3]$; the violet point is $\big(0, 0, f(0, 0)\big) = (0, 0, 0)$.

We prove that $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is a point of local minimum, but not a point of (global) minimum.

Example: points of local minimum (2/4)



Example: points of local minimum (3/4)

We use the spherical δ-neighborhood $N\left(\begin{pmatrix}0\\0\end{pmatrix}, 1\right)$.

For every $\begin{pmatrix}x_1\\x_2\end{pmatrix} \in N\left(\begin{pmatrix}0\\0\end{pmatrix}, 1\right)$ we have $x_2^2 \leq x_1^2 + x_2^2 < 1$, hence $1 - \frac{x_2^2}{2} > \frac{1}{2}$, and then:

$$f\left(\begin{pmatrix}x_1\\x_2\end{pmatrix}\right) - f\left(\begin{pmatrix}0\\0\end{pmatrix}\right) = x_1^2 + x_2^2 - \frac{1}{2}x_1^2 x_2^2$$

$$= x_1^2\left(1 - \frac{x_2^2}{2}\right) + x_2^2$$

$$\geq \frac{1}{2}x_1^2 + x_2^2 \geq 0,$$

with equality only for $x_1 = x_2 = 0$.

Example: points of local minimum (4/4)

It follows that

$$f\left(\begin{pmatrix}0\\0\end{pmatrix}\right) \leq f\left(\begin{pmatrix}x_1\\x_2\end{pmatrix}\right), \qquad \forall \begin{pmatrix}x_1\\x_2\end{pmatrix} \in N\left(\begin{pmatrix}0\\0\end{pmatrix}, 1\right),$$

and $\begin{pmatrix}0\\0\end{pmatrix}$ is a point of local minimum of f in $\mathbb{R}^2$, with local minimum 0.

It is not a point of (global) minimum, since

$$f\left(\begin{pmatrix}3\\3\end{pmatrix}\right) = 9 + 9 - \frac{81}{2} = -\frac{45}{2} < 0 = f\left(\begin{pmatrix}0\\0\end{pmatrix}\right).$$

Global and local optima

- Every point of (global) minimum is a point of local minimum (with any δ), and every point of (global) maximum is a point of local maximum.
- The converse is false: a point of local minimum may not be a point of minimum, as the previous example and the following one show.

Example: a local minimum that is not global (1/8)

Consider the following function:

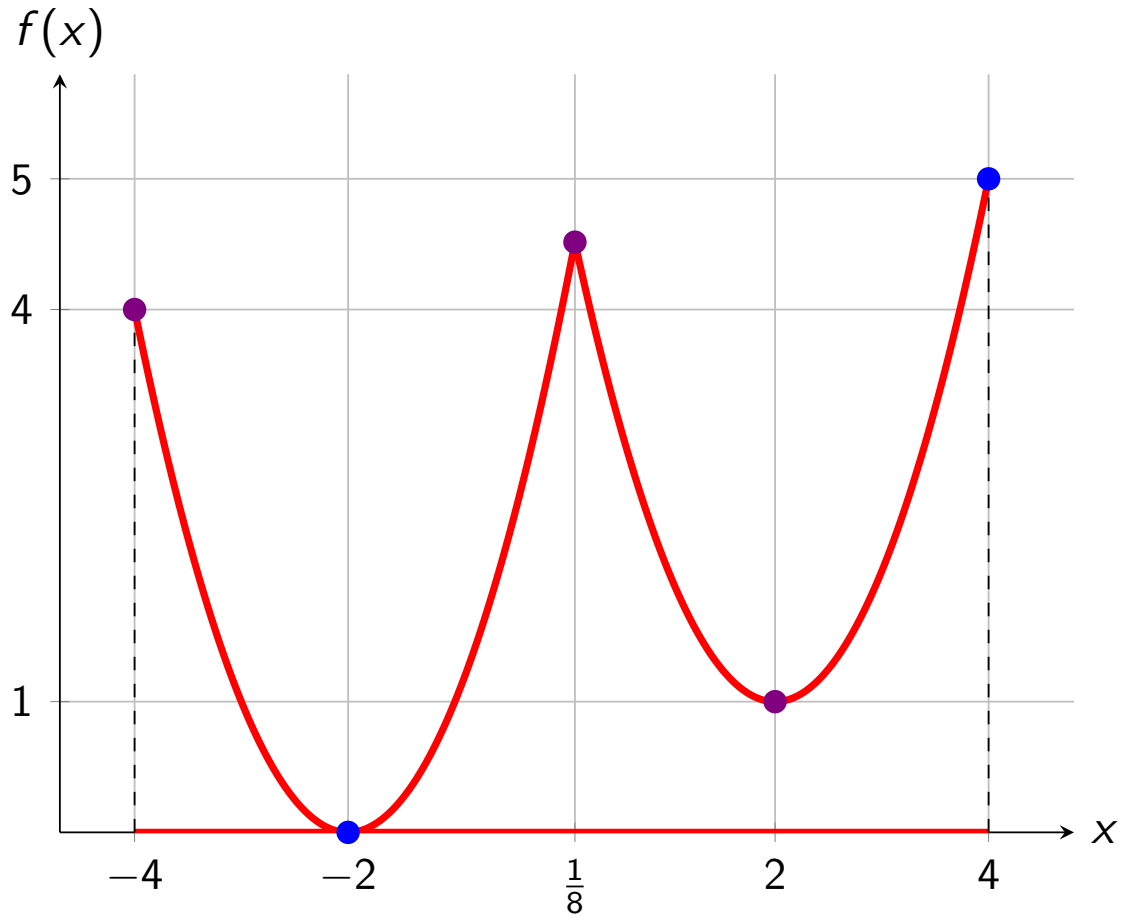
$$f : [-4, 4] \to \mathbb{R}, \; f : x \mapsto \min \left\{ (x+2)^2, \; (x-2)^2 + 1 \right\}.$$

Since $(x+2)^2 \leq (x-2)^2 + 1$ if and only if $8x \leq 1$, we have

$$f(x) = \begin{cases} (x+2)^2 & \text{if } -4 \leq x \leq \frac{1}{8}, \\ (x-2)^2 + 1 & \text{if } \frac{1}{8} \leq x \leq 4. \end{cases}$$

In the figure of the next slide, the points of global minimum and maximum are blue and the other points of local minimum and maximum are violet.

Example: a local minimum that is not global (2/8)



Example: a local minimum that is not global (3/8)

Global minimum.

- Both $(x+2)^2$ and $(x-2)^2+1$ are nonnegative, so $f(x) \geq 0 = f(-2)$ for every $x \in [-4, 4]$.
- Since $(x-2)^2+1 \geq 1$, $f(x) = 0$ only for $x = -2$.
- The point $x = -2$ is the unique point of minimum and the minimum is 0.

Example: a local minimum that is not global (4/8)

Local minimum that is not global.

- In the neighborhood $N\big((2),1\big) = (1,3)$ we have $x \geq \frac{1}{8}$, and then
$$f(x) = (x-2)^2 + 1 \geq 1 = f(2)$$
for every $x \in N\big((2),1\big) \cap [-4,4] = (1,3)$.
- The point $x = 2$ is a point of local minimum and 1 is a local minimum; it is not a point of (global) minimum, since $f(-2) = 0 < 1 = f(2)$.

Example: a local minimum that is not global (5/8)

Global maximum.

- For $-4 \leq x \leq \frac{1}{8}$ we have $-2 \leq x+2 \leq \frac{17}{8}$ and then

$$f(x) = (x+2)^2 \leq \left(\frac{17}{8}\right)^2 = \frac{289}{64} < 5.$$

- For $\frac{1}{8} \leq x \leq 4$ we have $-\frac{15}{8} \leq x-2 \leq 2$ and then

$$f(x) = (x-2)^2 + 1 \leq 4 + 1 = 5,$$

  with equality only for $x = 4$.
- The point $x = 4$ is the unique point of maximum and the maximum is 5.

Example: a local minimum that is not global (6/8)

Local maxima that are not global.

- In $N\big((-4),1\big) \cap [-4,4] = [-4,-3)$ we have $-2 \leq x+2 < -1$ and then
$$f(x) = (x+2)^2 \leq 4 = f(-4) :$$
the point $x = -4$ is a point of local maximum.
- In $N\big((\frac{1}{8}),1\big) = (-\frac{7}{8},\frac{9}{8})$:
  - when $x \leq \frac{1}{8}$, we have $0 < x+2 \leq \frac{17}{8}$, so that $f(x) = (x+2)^2 \leq \frac{289}{64}$;
  - when $x \geq \frac{1}{8}$, we have $-\frac{15}{8} \leq x-2 < 0$, so that
$$f(x) = (x-2)^2 + 1 \leq \frac{225}{64} + 1 = \frac{289}{64}.$$

Example: a local minimum that is not global (7/8)

- In both cases $f(x) \leq f(\frac{1}{8})$: the point $x = \frac{1}{8}$ is a point of local maximum.
- Neither $x = -4$ nor $x = \frac{1}{8}$ is a point of (global) maximum, since

$$f(-4) = 4 < 5 \qquad \text{and} \qquad f\left(\frac{1}{8}\right) = \frac{289}{64} < 5.$$

Example: a local minimum that is not global (8/8)

Consider the optimization problem

$$\min\{\, f(x) \text{ subject to } -4 \leq x \leq 4 \,\}.$$

- A method that only looks at the points close to $x = 2$ finds the local minimum 1.
- But the optimal value is $z(P) = 0$, attained at $\tilde{\boldsymbol{x}}(P) = (-2)$.

Summary

Global optima:

- Points of minimum (maximum) of f in D: $f(\tilde{\boldsymbol{x}}) \leq f(\boldsymbol{x})$ $(\geq)$ for every $\boldsymbol{x} \in D$
- The optimal solutions of a problem are the global optima of f in F

Local optima:

- Spherical δ-neighborhood: $N(\bar{\boldsymbol{x}}, \delta) = \{\boldsymbol{x} : ||\bar{\boldsymbol{x}} - \boldsymbol{x}|| < \delta\}$
- Point of local minimum (maximum): the inequality holds only in $N(\bar{\boldsymbol{x}}, \delta) \cap D$, for one $\delta > 0$
- Every global optimum is a local optimum; the converse is false

Key messages

Summary

1. **Optimization problems:**
   - Find $\tilde{\boldsymbol{x}}(P) \in F$ that minimizes (maximizes) f; $z(P)$ is the optimal value
   - No optimal solution: unbounded, infeasible, unattainable infimum (supremum)
   - Every feasible solution gives a bound on $z(P)$

2. **Properties:**
   - $\min\{f\} = -\max\{-f\}$; adding constants and multiplying by positive constants does not change the optimal solutions

3. **Local and global optima:**
   - Every global optimum is local, but a local optimum may not be global